

Kumudwati Kakani

Charlotte, North Carolina

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Summary

Data Analyst with over 4 years of experience designing and implementing data-driven solutions across healthcare and manufacturing sectors. Proficient in Python, SQL, AWS (S3, Redshift, Lambda, SageMaker), and modern data visualization tools. Demonstrated expertise in developing end-to-end data pipelines, integrating predictive models, and streamlining analytical workflows. Adept in Agile methodologies, SDLC, and collaborative problem-solving to support strategic decision-making in regulated, high-impact environments.

Technical Skills

Programming Languages: SQL, ANSI SQL, Python, R, JavaScript, Shell Scripting, HTML/CSS

Technologies & Platforms: AWS (S3, Redshift, SageMaker, Rekognition, Cloud Compliance), Apache Spark ALS, Snowflake, Talend Open Studio, Alteryx, Trifacta, Informatica Data Quality, Google BigQuery, Amazon Kinesis, ElasticSearch, NLP (Basic), HIPAA-Compliant Environments

Data Visualization & Reporting: Tableau, Power BI, Streamlit, Tableau Prep, Jupyter Notebooks, Microsoft Excel

Libraries & Tools: Git, Docker, SharePoint, Confluence, VS Code, SQL Workbench, Colibra, Ataccama, OpenRefine, Spark Streaming

Soft Skills: Analytical Thinking, Cross-functional Collaboration, Data Storytelling, Attention to Detail, Time Management, Problem Solving, Adaptability, Workflow Optimization, Agile Communication, Regulatory Awareness

Work Experience

University of North Carolina at Charlotte, Charlotte, NC

Jan 2024 – May 2025

Graduate Assistant – Healthcare Data Analysis

- Assisted Dr. Timothy Lowe as a Data Analyst for a graduate-level course in healthcare data analytics, supporting instruction and analysis of public health datasets.
- Engineered and deployed scalable, fault-tolerant distributed systems to support high-throughput data storage, indexing, and querying, reducing latency and operational overhead across multiple services.
- Developed predictive models and computation frameworks to run real-time inferences on distributed platforms, improving system responsiveness and data-driven decision-making at scale.
- Led the design and implementation of object-oriented software solutions for complex, broadly defined problems using languages such as Java and Python, improving code reusability and maintainability.
- Delivered high-quality, agile-based software iterations by participating in sprint planning, code reviews, and continuous integration pipelines, enhancing development velocity and team collaboration.
- Researched and optimized complex algorithms and data structures to improve performance in multi-tiered architectures, contributing to cost reduction and processing efficiency.
- Streamlined problem-solving strategies in ambiguous environments by applying strong fundamentals in computer science and optimization techniques, enabling quicker prototyping and system stability.
- Designed and delivered analytics demonstrations using SAS, R, SQL, Python, and Tableau, helping students build practical skills in data analysis and visualization.
- Evaluated technical assignments and team projects for accuracy and alignment with MPH/CEPH standards.
- Mentored students in applying statistical methods, interpreting data, and creating impactful dashboards.
- Developed reusable Jupyter Notebook templates and self-paced resources to support instructional efficiency.
- Supported instruction under Prof. Richard Sheinis, Esq. in a graduate course covering HIPAA, healthcare cybersecurity, and legal frameworks.
- Analyzed real-world legal use cases involving patient data privacy, ethical handling, and compliance risk.
- Provided academic support on HIPAA quizzes, regulatory assignments, and midterm evaluations.
- Assisted students in understanding healthcare data governance, U.S. privacy regulations, and cybersecurity standards through Zoom and Piazza.
- Ensured instructional content aligned with real-world regulatory expectations and best practices.

Tech Stack: SAS, R, SQL, Python, Tableau, Jupyter Notebooks, Microsoft Excel, Canvas, Zoom, Piazza, Confluence, HIPAA Compliance Tools, Data Governance Frameworks.

Metalyst Forgings Limited, India

Feb 2020 to June 2023

Data analyst

- Conducted complex data querying and management using MY SQL and ANSI SQL, supporting critical data operations.
- Designed and managed databases using MySQL Workbench, improving data structure and accessibility.
- Delivered actionable insights by analyzing cross-functional datasets to support strategic initiatives.
- Developed real-time dashboards in Power BI and Tableau, enabling data-driven decision-making across teams.
- Ensured data quality and integrity through automated SQL validation and error-checking processes.
- Partnered with remote, cross-functional teams to translate business needs into scalable data solutions.
- Communicated complex trends through clear visualizations and executive-level reports.
- Integrated cloud-based data sources (e.g., AWS S3, Redshift, BigQuery) for streamlined analytics workflows.
- Applied statistical analysis and forecasting to uncover trends and optimize business performance.
- Advanced data manipulation and analysis performed using Microsoft Excel, supporting key business decisions.

- Created dynamic data visualizations and reports using Tableau, enhancing understanding of data patterns.
- Managed data integration projects effectively using Talend Open Studio, ensuring data consistency and reliability.
- Utilized GIT for version control, enhancing data security and collaboration among project teams.
- Employed Docker to ensure consistent development environments across various projects, reducing setup times.
- Analyzed and interpreted complex datasets using Jupyter Notebooks, driving insights and analytics initiatives.
- Utilized Microsoft Excel for financial modeling and forecasting, providing detailed analysis and reporting.
- Applied GIT for effective version control in collaborative development projects, maintaining code integrity.
- Leveraged Docker for containerization, ensuring seamless environment management for development teams.
- Used Jupyter Notebooks for data visualization and predictive modeling, enhancing data-driven decisions.
- Developed ETL processes with Talend Open Studio, improving data integration and workflow efficiency.
- Employed Tableau for strategic data reporting, supporting business intelligence and decision-making processes.
- Managed version control with GIT in development projects, ensuring efficient team collaboration and data consistency.

Tech Stack: MySQL, ANSI SQL, MySQL Workbench, Microsoft Excel, Tableau, Talend Open Studio, GIT, Docker, Jupyter Notebooks

Education

- **Masters in Health Informatics and Analytics** | The University of North Carolina at Charlotte.
- **B. Tech in Biotechnology** (Major - Bio Informatics) | KL University

Certifications

- AWS Certified Data Engineer - Associate

Projects

- **ICU Length of Stay Prediction with Explainability**

Python, Random Forest, SHAP, LIME, Flask

Developed and deployed a predictive analytics web application to forecast ICU length of stay using Random Forest models. Integrated SHAP and LIME explainability for transparent decision-making by clinicians. Streamlined front-end and back-end data collection workflows while ensuring alignment with healthcare compliance standards.

- **Healthcare Revenue Cycle Automation via FHIR API**

Python, FHIR API, Snowflake, BigQuery, ETL Automation

Engineered scalable, HIPAA-compliant ETL pipelines to automate claims and insurance data ingestion using FHIR APIs. Integrated structured data into Snowflake and BigQuery, reducing claims cycle time by 40% and improving financial reporting accuracy and operational efficiency.

- **Omnichannel Risk Detection for Chronic Diseases**

Python, Machine Learning, EHR Data, Tableau, Streamlit

Designed and implemented machine learning models to predict risk factors for heart disease, diabetes, and cancer using multi-source EHR datasets. Developed interactive dashboards in Tableau and Streamlit to support proactive care management and targeted health interventions.

- **EHR Interoperability and Privacy-Driven Data Exchange Optimization**

Python, Metadata Lineage, HIPAA Compliance, Data Governance

Led a system-wide audit of EHR interoperability gaps and implemented metadata lineage tracking and privacy-preserving techniques to enhance secure data exchange. Ensured compliance with HIPAA and internal governance frameworks, improving data quality and trust in healthcare data-sharing processes.